# MSG No. 69.521 $Fmmm$ [ Type I, orthorhombic ]

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

| No. | tag | matrix (polar) | det | TR |
|---|---|---|---|---|
| 1 | $\{1\|0\}$ | $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ | 1 | 1 |
| 2 | $\{2_{100}\|0\}$ | $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$ | 1 | 1 |
| 3 | $\{2_{010}\|0\}$ | $\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$ | 1 | 1 |
| 4 | $\{2_{001}\|0\}$ | $\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ | 1 | 1 |
| 5 | $\{-1\|0\}$ | $\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$ | −1 | 1 |
| 6 | $\{m_{100}\|0\}$ | $\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ | −1 | 1 |
| 7 | $\{m_{010}\|0\}$ | $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ | −1 | 1 |
| 8 | $\{m_{001}\|0\}$ | $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$ | −1 | 1 |
| 9 | $\{1\|0\frac{1}{2}\frac{1}{2}\}$ | $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$ | 1 | 1 |
| 10 | $\{2_{100}\|0\frac{1}{2}\frac{1}{2}\}$ | $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$ | 1 | 1 |
| 11 | $\{2_{010}\|0\frac{1}{2}\frac{1}{2}\}$ | $\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$ | 1 | 1 |
| 12 | $\{2_{001}\|0\frac{1}{2}\frac{1}{2}\}$ | $\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$ | 1 | 1 |
| 13 | $\{-1\|0\frac{1}{2}\frac{1}{2}\}$ | $\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$ | −1 | 1 |

*continued ...*

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| No. | tag | matrix (polar) | det | TR |
|---|---|---|---|---|
| 14 | $\{\mathrm{m}_{100}\|0\frac{1}{2}\frac{1}{2}\}$ | $\begin{bmatrix}-1&0&0&0\\0&1&0&\frac{1}{2}\\0&0&1&\frac{1}{2}\end{bmatrix}$ | $-1$ | 1 |
| 15 | $\{\mathrm{m}_{010}\|0\frac{1}{2}\frac{1}{2}\}$ | $\begin{bmatrix}1&0&0&0\\0&-1&0&\frac{1}{2}\\0&0&1&\frac{1}{2}\end{bmatrix}$ | $-1$ | 1 |
| 16 | $\{\mathrm{m}_{001}\|0\frac{1}{2}\frac{1}{2}\}$ | $\begin{bmatrix}1&0&0&0\\0&1&0&\frac{1}{2}\\0&0&-1&\frac{1}{2}\end{bmatrix}$ | $-1$ | 1 |
| 17 | $\{1\|\frac{1}{2}0\frac{1}{2}\}$ | $\begin{bmatrix}1&0&0&\frac{1}{2}\\0&1&0&0\\0&0&1&\frac{1}{2}\end{bmatrix}$ | 1 | 1 |
| 18 | $\{2_{100}\|\frac{1}{2}0\frac{1}{2}\}$ | $\begin{bmatrix}1&0&0&\frac{1}{2}\\0&-1&0&0\\0&0&-1&\frac{1}{2}\end{bmatrix}$ | 1 | 1 |
| 19 | $\{2_{010}\|\frac{1}{2}0\frac{1}{2}\}$ | $\begin{bmatrix}-1&0&0&\frac{1}{2}\\0&1&0&0\\0&0&-1&\frac{1}{2}\end{bmatrix}$ | 1 | 1 |
| 20 | $\{2_{001}\|\frac{1}{2}0\frac{1}{2}\}$ | $\begin{bmatrix}-1&0&0&\frac{1}{2}\\0&-1&0&0\\0&0&1&\frac{1}{2}\end{bmatrix}$ | 1 | 1 |
| 21 | $\{-1\|\frac{1}{2}0\frac{1}{2}\}$ | $\begin{bmatrix}-1&0&0&\frac{1}{2}\\0&-1&0&0\\0&0&-1&\frac{1}{2}\end{bmatrix}$ | $-1$ | 1 |
| 22 | $\{\mathrm{m}_{100}\|\frac{1}{2}0\frac{1}{2}\}$ | $\begin{bmatrix}-1&0&0&\frac{1}{2}\\0&1&0&0\\0&0&1&\frac{1}{2}\end{bmatrix}$ | $-1$ | 1 |
| 23 | $\{\mathrm{m}_{010}\|\frac{1}{2}0\frac{1}{2}\}$ | $\begin{bmatrix}1&0&0&\frac{1}{2}\\0&-1&0&0\\0&0&1&\frac{1}{2}\end{bmatrix}$ | $-1$ | 1 |
| 24 | $\{\mathrm{m}_{001}\|\frac{1}{2}0\frac{1}{2}\}$ | $\begin{bmatrix}1&0&0&\frac{1}{2}\\0&1&0&0\\0&0&-1&\frac{1}{2}\end{bmatrix}$ | $-1$ | 1 |
| 25 | $\{1\|\frac{1}{2}\frac{1}{2}0\}$ | $\begin{bmatrix}1&0&0&\frac{1}{2}\\0&1&0&\frac{1}{2}\\0&0&1&0\end{bmatrix}$ | 1 | 1 |
| 26 | $\{2_{100}\|\frac{1}{2}\frac{1}{2}0\}$ | $\begin{bmatrix}1&0&0&\frac{1}{2}\\0&-1&0&\frac{1}{2}\\0&0&-1&0\end{bmatrix}$ | 1 | 1 |
| 27 | $\{2_{010}\|\frac{1}{2}\frac{1}{2}0\}$ | $\begin{bmatrix}-1&0&0&\frac{1}{2}\\0&1&0&\frac{1}{2}\\0&0&-1&0\end{bmatrix}$ | 1 | 1 |

*continued ...*

Table 1

| No. | tag | matrix (polar) | det | TR |
|---|---|---|---|---|
| 28 | $\{2_{001}\vert\frac{1}{2}\frac{1}{2}0\}$ | $\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$ | 1 | 1 |
| 29 | $\{-1\vert\frac{1}{2}\frac{1}{2}0\}$ | $\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & 0 \end{bmatrix}$ | −1 | 1 |
| 30 | $\{\mathrm{m}_{100}\vert\frac{1}{2}\frac{1}{2}0\}$ | $\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$ | −1 | 1 |
| 31 | $\{\mathrm{m}_{010}\vert\frac{1}{2}\frac{1}{2}0\}$ | $\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$ | −1 | 1 |
| 32 | $\{\mathrm{m}_{001}\vert\frac{1}{2}\frac{1}{2}0\}$ | $\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & 0 \end{bmatrix}$ | −1 | 1 |